SC2014 : Sensors, Interfacing & Digital Control

Part 1 : Tutorial X – Analogue-Digital Filters

Q1 : A cable-based sensing system is used to transmit/receive signal from the remote transducer/sensor. The sensor signal is often corrupted by some high frequency noise. Suppose the signal data has a frequency spectrum of 1 kHz to 2 kHz and the high-frequency noise has a frequency spectrum of 80 kHz to 100 KHz.

Design a suitable *RC* (passive) filter to extract the signal at the receiving end such that the signal gain is maintain at 1. Assume that 0.1 µF capacitors and 0-10 kΩ variable resistors are available for the design.

Q2 : The standard passive filter characteristic is shown below. Identify what is the filter type for this Laplace Equation.

$$H(s) = \frac{s}{s + \omega_c}$$

Design the suitable *RC* (passive) filter for a cut-off frequency of 500Hz and its component value. You can assume that 0.1 µF capacitors and 0-10 kΩ variable resistors are available for the design.

Q3 : We can design a Digital Filter based on the Analogue Filter specifications of **Q2**.

$$H(s) = \frac{s}{s + \omega_c} = \frac{s}{s + 2\pi(500)}$$

Using bilinear transformation and a sampling frequency of 5000 samples/sec, derive the difference equation (input-output equation).

Q4 : From Tut2 Question2, you have derived the difference equations required for implementing a Low-Pass filter. This is achieved via a Numerical Method known as Backward Euler Method. The solutions includer different bandwidths but same sampling rate are (shown below):

Sampling = 1000 Samples/sec, Bandwidth 10Hz

$$V_o[n] = (0.06)V_{in}[n] + (0.94)V_o[n-1]$$

Sampling = 1000 Samples/sec, Bandwidth 100Hz

$$V_o[n] = (0.38)V_{in}[n] + (0.62)V_o[n-1]$$

With the Low-Pass 1st Order Low-Pass Digital Filter (via Bilinear Transformation) example. You can also work out the difference equations for the Digital Filter Implementation at the same sampling rate of 1000 sample/sec and for two different bandwidth frequency of 10Hz & 100Hz.

Compare and contrast the difference between the coefficients/terms of the difference equation (input-output equation) and state the pros & cons of the two methods.